

Question	Answer	Mark
2(a)(i)	(work =) force $\times$ distance moved in the direction of the force	B1
2(b)(i)	$\rho = m/V$	C1
	$= (20/9.81)/(4/3 \times \pi \times 0.16^3)$	A1
	$= 120 \text{ kg m}^{-3}$	
2(b)(ii)	the pressure on the lower surface (of sphere) is greater than the pressure on the upper surface (of sphere)	B1
2(b)(iii)	$a = (170 - 20)/(20/9.81)$	C1
	$= 74 \text{ ms}^{-2}$	A1
2(b)(iv)	$D = 170 - 20 (= 150)$	C1
	$810 \times (0.16^2) \times v^2 = 150$	C1
	$v = 2.7 \text{ ms}^{-1}$	A1
2(b)(v)	$4870 = (4850 \times v)/(v - 6.30)$	C1
	$v = 1530 \text{ ms}^{-1}$	A1

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2(c)	15.41 g	M1